



Refining the Sunspot Number Series to ISN V3

CHALLENGES AND BENEFITS FOR THE SPACE CLIMATE COMMUNITY

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Mission Brief

FARSuN Project

Why FARSuN?

The International Sunspot Number (ISN) underpins space climate diagnostics, yet two major recalibrations still leave observer-dependent jumps (e.g. the 1947 "Waldmeier break").

FARSuN (Findability and Accessibility of historical Raw Sunspot Numbers) is rebuilding the index from primary sources (1607–1980) with traceable metadata, uncertainties, and open delivery.

Mission Recover and standardise dispersed sunspot archives (drawings, tables, weather notes).

Outcome Deliver ISN V3 with hemispheric indices, provenance, and FAIR-compliant access.

Project Pulse (Early 2025)

1607–1980

Temporal coverage of curated sources

95% of 2 000+

Zürich observation tables digitised

(Transkribus + manual QC)

1 056

Entries from C. H. Adams notebooks

with spotless-day tagging

100%

Mittheilungen volumes integrated

into database

400+

Unique historical observers linked to

standard IDs

Ready for pilot

Citizen-science portal blueprint and

QA workflow

Source: FARsUN interactive report (*'site/webpage1.html'*) and dataset QMD dossiers.

Why Reconstruction

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Limitations of Legacy Recalibration

Symptoms

- Residual scale jumps ("Waldmeier jump", late-19th c. transitions) in SN V2 derived records.
- Missing metadata: observer aliases, instrument notes, weather context drop out of composite series.
- Uncertainties opaque; difficult to quantify spotless-day reliability or observer drift.

Why Tweaking Coefficients Is Not Enough

- Linear rescaling keeps biases embedded in the stitched index.
- Downstream models inherit hidden assumptions, limiting predictive skill.
- Science questions (hemispheric asymmetry, climate forcing) demand provenance-rich data.

Reconstruction Advantages

Legacy recalibration (SN V2)

- Adjusts existing composite, no new metadata.
- Biases from observer transitions persist.
- Limited uncertainty audit trail.

FARSuN reconstruction (ISN V3)

- Revisits original artefacts (drawings, logbooks, weather notes).
- Builds calibration curves per observer using raw counts.
- Publishes machine-readable products with explicit uncertainties and provenance.

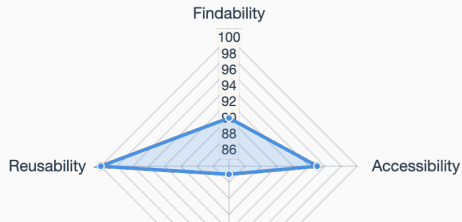
Narrative adapted from the "Problem" and "Solution" sections in '*site/webpage1.html*'.

Methodology

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Four-Axis Workflow

1. **Gather** Locate and digitise dispersed historical sources (Wolf's *Mittheilungen*, Zürich tables, Gruithuisen ledgers, Adams drawings).
2. **Process** Blend handwriting/OCR engines (Transkribus) with manual keying for fragile tables and drawings.
3. **Validate** Reconcile observer IDs, calendars, and counting conventions; propagate spotless-day confidence levels.
4. **Disseminate** Release FAIR-ready datasets, APIs, and VO-compliant services with versioned uncertainties.



Quality Gates and Tooling

Verification

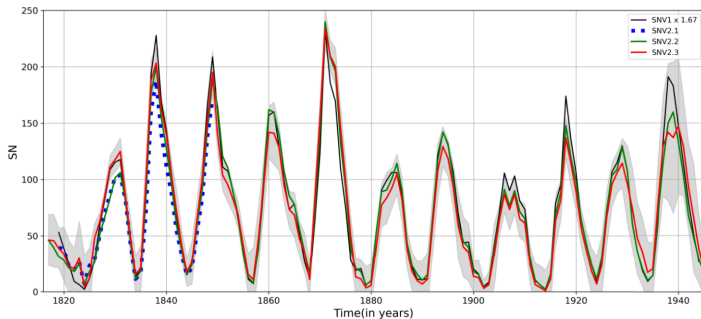
- Dual review of Adams and Gruithuisen counts before ingest.
- Cross-matching against Wolf and SILSO indices to flag drifts.
- Automated scripts check metadata completeness and naming standards.

Toolchain

- Transkribus handwriting models + manual correction interface.
- Python pipelines (Git-tracked) orchestrating OCR, QC, and format exports.
- Planned citizen science portal with training sets for Kurrent transcription.

Data Pipeline to ISN V3

1. Curate daily group/spot counts with provenance-rich metadata.
2. Solve observer response curves to harmonise scales over centuries.
3. Aggregate to monthly, yearly, and hemispheric indices with propagated uncertainties.
4. Benchmark against SN V2, satellite-era proxies, and reconstructions to validate corrections.



Historical Assets

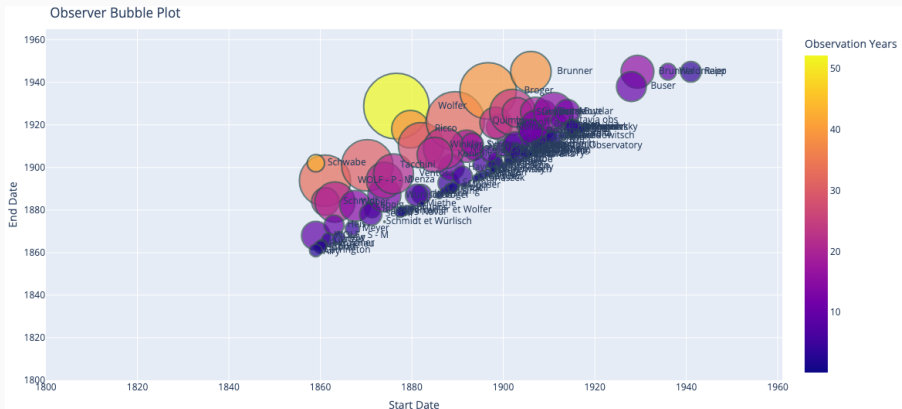
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Key Historical Collections

- **Zürich Tables (1945–1979):** 95% digitised; metadata for late-1970s observers underway.
- **Wolf's *Mittheilungen* (1610–1918):** fully integrated with observer statistics over 400+ years.
- **Gruithuisen Manuscripts (1817–1849):** 25% transcribed; bespoke renaming + QC scripts for Kurrent entries.
- **C. H. Adams Drawings (1819–1823):** 338 sketches extracted, 1 056 entries including spotless-day tags.
- **Augustin Stark Ledgers (1813–1835):** harmonising tables + weather annotations.
- **Rost Journals (18th–19th c.):** metadata vocabulary build prior to digitisation ingest.
- **Supporting collections:** Gruithuisen weather logs, Adams correspondence, Zürich observer notes.



Observer Coverage



Bubble size = total observations; colour encodes observing years (source: '[site/webpage1.html](#)').

Science Impact

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Addressing Core Challenges

Data hurdles

- Deciphering 19th-century *Kurrent* handwriting and marginalia.
- Normalising heterogeneous formats (drawings, tables, narrative notes).
- Tracking provenance from scanned folio to released dataset.

Mitigations

- Specialist training, handwriting models, and dual QC.
- Schema mapping + metadata vocabularies across archives.
- Git-backed pipelines with audit trails and uncertainty flags.

Benefits for the Community

Forecasting Bias-reduced ISN V3 improves solar cycle prediction skill and operational space weather baselines.

Climate Validated long-term forcing records refine Earth system and climate models.

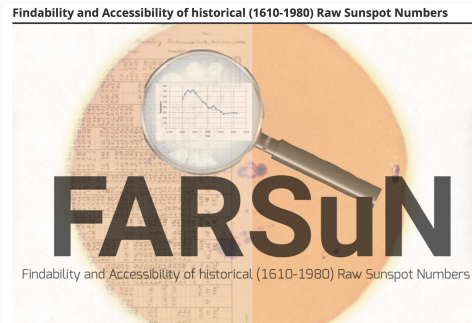
Citizen science Public transcription portal accelerates data recovery and outreach. **Open science** FAIR datasets, APIs, and reproducible notebooks catalyse external research.

Path Ahead

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2025 Roadmap

- Launch citizen-science transcription pilot with curated training sets.
- Finalise Zürich metadata encoding and close remaining digitisation gaps.
- Publish ISN V3 release notes with full uncertainty budget and API endpoints.
- Expand collaborations (ROB, Nagoya, UCLouvain) for cross-validation and joint publications.



Q&A



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Questions?

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